MODULE 33 · SECTION 5 - MAINTAINING AND AUTOMATING DATA WORKLOADS

BigQuery Capacity: On-Demand, Editions and Reservations

On-demand or capacity, the three editions, baseline and autoscaling slots, commitments, assignments, idle slots, priority and queues, and cost guard rails.

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On-demand bills the bytes you read.
Capacity bills the slots you hold. Choose the bill you can predict.

On-demand or capacity?

ON-DEMAND

Pay per byte

- Billed per TiB processed, first TiB a month free
- About 2,000 slots per project, with bursts
- No commitments, nothing to manage
- Custom quotas cap the bytes

CAPACITY

Pay per slot

- Slots billed per second, one-minute minimum
- Reservations, autoscaling, commitments
- Spend capped by reservation size
- Pick an edition: Standard, Enterprise, Plus

THE TIEBREAKER

Small, spiky use: on-demand. Steady heavy use, or a bill that must not surprise: capacity.

Three editions, compared

EDITION	CAPACITY	ADDS
Standard	Autoscaling only, up to 1,600 slots, no commitments	The basics: no BigQuery ML, no BI Engine, no indexes
Enterprise	Baseline plus autoscaling, with commitments	BigQuery ML, BI Engine, search and vector indexes, continuous queries
Enterprise Plus	As Enterprise	Managed disaster recovery, Assured Workloads compliance

HONEST LIMIT

Google warns that features move between editions. Choose for the workload, then check the current table.

Baseline and autoscaling

- A **slot** is a unit of compute: CPU, memory and I/O together.
- **Baseline** slots are always yours, and always billed, used or not.
- **Autoscaling** adds slots in steps of 50, up to the reservation's maximum.
- Scaled slots are kept for **60 seconds** or more, and billed whether used or not.

> CURRENCY CHECK
>
> Fluid scaling, which you opt into, removes the one-minute minimum for bursty work.

Committing for a discount

- **One-year** or **three-year** commitments, bought in steps of 50 slots.
- They cover **baseline** slots only. Autoscaled slots pay the standard rate.
- They cannot be cancelled early. Set renewal to NONE to stop them renewing.
- Billed to the **administration project**. Unused committed slots become idle slots.

> THE TRAP
>
> Standard edition has no commitments. A question that pairs them is a trap.

Larkspur's reservations, in SQL

```
1 CREATE RESERVATION `lk-admin-prod.region-europe-west2.etl`
2 OPTIONS (edition = 'ENTERPRISE', slot_capacity = 300, autoscale_max_slots = 700);
3
4 CREATE ASSIGNMENT `lk-admin-prod.region-europe-west2.etl.ingest_queries`
5 OPTIONS (assignee = 'projects/lk-ingest-prod', job_type = 'QUERY');
6
7 CREATE RESERVATION `lk-admin-prod.region-europe-west2.bi`
8 OPTIONS (edition = 'ENTERPRISE', slot_capacity = 0, autoscale_max_slots = 400);
```

2 300 baseline slots for the pipelines, autoscaling to 700 at most.

8 No baseline for analysts: pay only when they query, capped at 400.

FOR LARKSPUR

The maximum is the spending ceiling. That is what the finance director asked for.

Who uses which reservation

- Assign a reservation to an **organisation**, **folder** or **project**. The most specific wins.
- Job types: **QUERY**, **PIPELINE**, **BACKGROUND**, ML_EXTERNAL, **CONTINUOUS**.
- No assignment means **on-demand**. An assignment to **None** forces on-demand.
- Loads are free on a shared pool. Assigning PIPELINE moves them onto your paid slots.

> THE TIEBREAKER
>
> Opting one project out of an organisation-wide reservation: assign it to None.

Idle slots and fairness

- Reservations borrow **idle slots** from each other, within one admin project and edition.
- Borrowed slots go back within milliseconds when the owner needs them.
- **ignore_idle_slots** stops a reservation borrowing. It can still lend.
- **Fair scheduling** splits a reservation's slots between the projects using it.

> HONEST LIMIT
>
> Never plan a deadline on idle slots. They can be taken back at any moment.

Interactive, batch and queues

- BigQuery sets **concurrency** dynamically. Extra queries wait in a queue.
- Queues hold **1,000 interactive** and **20,000 batch** queries, per project per region.
- Queued queries time out after 6 hours, or 24 hours for batch, by default.
- **Batch priority**: a separate cap, and interactive queries go first.

> THE TIEBREAKER
>
> Reports that can finish any time before morning: batch priority.

Capping cost on-demand

- **QueryUsagePerDay** caps a project. Default: 200 TiB a day.
- **QueryUsagePerUserPerDay** caps each user in a project. Default: unlimited.
- Checked before a query runs, approximate, and reset at **midnight Pacific**.
- **Maximum bytes billed** stops one query, from module 28. Quotas cap a day.

> REMEMBER
>
> Custom quotas apply to on-demand only. On capacity, the reservation's maximum is the cap.

Larkspur's capacity, end to end

01

lk-admin-prod

Holds the commitments and reservations.

02

etl reservation

Baseline 300, up to 700. Pipelines.

03

bi reservation

No baseline, up to 400. Analysts.

04

Sandboxes

Assigned to None. On-demand with quotas.

THE PATTERN

A committed baseline for steady work, a ceiling on everything else: a bill that can be forecast.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
Spend must be **predictable**	Reservation, with a maximum size
Steady load, **cheapest** long term	Commitment covering the baseline
Needs **BigQuery ML** or **BI Engine** on capacity	Enterprise edition
Managed DR, **FedRAMP** or ITAR	Enterprise Plus
One project **out of** the org reservation	Assign it to None
Can finish **any time overnight**	Batch priority

THE PATTERN

What must be predictable, what must be fast, and what can wait: size the bill around those.

> After one on-demand month cost GBP 40,000, the finance director wants a BigQuery bill that cannot exceed a set monthly figure.
>
> Pipelines run steadily all day. Analyst queries come in bursts.

What should the team set up?

A Custom per-user quotas on every project, staying on-demand

B On-demand, with maximum bytes billed set on every query

C Enterprise reservations: a committed baseline, autoscaling capped

D Standard edition, with autoscaling and no maximum set on any reservation

◆ PAUSE AND DECIDE

What should the team set up?

A Per-user quotas

Cap bytes, but the total still moves, and work stops at the cap.

B Maximum bytes billed

Caps each query, not the month.

C Committed baseline, capped

Steady work is discounted. Bursts stop at the ceiling.

D Standard, uncapped

No commitments, and no ceiling either.

THE SIGNAL

A bill that cannot exceed a figure: reservations with a maximum, and a committed baseline.

> Larkspur is moving BigQuery to capacity pricing. The team needs BigQuery ML, BI Engine and vector indexes for the field engineer assistant.
>
> It has no need for managed disaster recovery or FedRAMP-style compliance controls.

Which edition fits best?

A Enterprise edition, baseline plus autoscaling

B Standard edition, which only autoscales

C Enterprise Plus, with every feature switched on

D Standard edition, with a three-year commitment

◆ PAUSE AND DECIDE

Which edition fits best?

A **Enterprise**
BigQuery ML, BI Engine and indexes, without paying for Plus.

B **Standard**
No BigQuery ML, no BI Engine, no indexes.

C **Enterprise Plus**
Works, but pays for DR and compliance nobody needs.

D **Standard with commitment**
Standard offers no commitments at all.

THE SIGNAL

BigQuery ML, BI Engine and indexes, without DR or compliance needs: Enterprise.

> An organisation-wide reservation covers every Larkspur project. Analysts' experiments in sandbox projects now take slots the ETL pipelines need.
>
> Sandbox work should not touch the reservation at all, but must stay cheap.

What should the team do?

A. Delete the organisation-wide assignment, for every project

B. Buy more committed slots, so that there is room for both

C. Set ignore_idle_slots on the ETL reservation

D. Assign the sandbox projects to None, and add custom quotas

◆ PAUSE AND DECIDE

What should the team do?

A Delete the assignment

Pushes every project, pipelines included, to on-demand.

B Buy more slots

Pays more, and sandboxes still share the reservation.

C ignore_idle_slots

Stops borrowing, not sandboxes using the same reservation.

D Assign to None

Sandboxes go on-demand, and quotas keep them cheap.

THE SIGNAL

One set of projects out of an org reservation: assign them to None.

> Hundreds of report queries start at 01:00 each night. Some fail with queue errors, and the rest slow early interactive work.
>
> The reports only need to be ready by 07:00.

What should the team change?

A Raise the reservation's autoscaling maximum for the night

B Run the report queries with batch priority

C Spread the reports across more projects, to avoid the queue limit

D Set the queue timeout to -1, so that waiting queries fail quickly

◆ PAUSE AND DECIDE

What should the team change?

A Raise the maximum

Pays more for work that can simply wait.

B Batch priority

A queue of 20,000, and interactive work goes first.

C More projects

Dodges limits, and complicates billing and access.

D Timeout of -1

Turns queuing off, so even more queries fail.

THE SIGNAL

Work that can wait until morning: batch priority.

> A reservation has no baseline and autoscales to 2,000 slots. Hundreds of short, bursty queries run each hour.
>
> Monitoring shows about 300 slots in use on average, yet the bill matches far more.

What best explains the bill?

A Scaled slots bill for a minute or more, whether used or not

B Idle slots borrowed from other reservations are billed twice

C Commitments are charged again for every autoscaled slot

D Short queries cannot use cached results under capacity pricing

◆ PAUSE AND DECIDE

What best explains the bill?

A Scaled, not used

Each burst scales up, and pays for a minute or more.

B Idle slots billed twice

Borrowed idle slots are not billed to the borrower again.

C Commitments on autoscaling

Commitments cover baseline only.

D No cache

Cache rules do not change with pricing model.

THE SIGNAL

Bill above usage on bursty autoscaling: scaled slots, minute minimum. Try fluid scaling or a baseline.

> Larkspur bought a three-year commitment of 500 slots. Every reservation has a baseline of zero, autoscales to 800, and has ignore_idle_slots set to true.
>
> The bill shows the commitment, plus full-price autoscaling on top.

What should the team change?

A Cancel the commitment, and rely on autoscaling slots alone

B Set reservation baselines to use the 500 committed slots

C Raise every reservation's autoscaling maximum to 1,300

D Move every reservation down to the Standard edition

◆ PAUSE AND DECIDE

What should the team change?

A Cancel

Commitments cannot be cancelled early.

B Baselines on the commitment

Committed slots finally do the work, at the discount.

C Raise the maximum

More full-price autoscaling.

D Move to Standard

No commitments, and features are lost.

THE SIGNAL

Paying a commitment and autoscaling together: the commitment must cover baselines.

Module 33 in five lines

- On-demand bills **bytes**. Capacity bills **slots**, and its maximum caps the bill.
- **Standard** is basic, **Enterprise** adds ML and BI Engine, **Plus** adds DR and compliance.
- Autoscaling bills **scaled** slots. Commitments cover **baseline** only.
- Assignments by job type. **None** opts a project out. Idle slots are shared.
- **Batch priority** for work that waits. Custom quotas cap on-demand days.

COMING UP

Next: module 34, right-sizing clusters and pipelines.